
TECHNICAL BULLETIN

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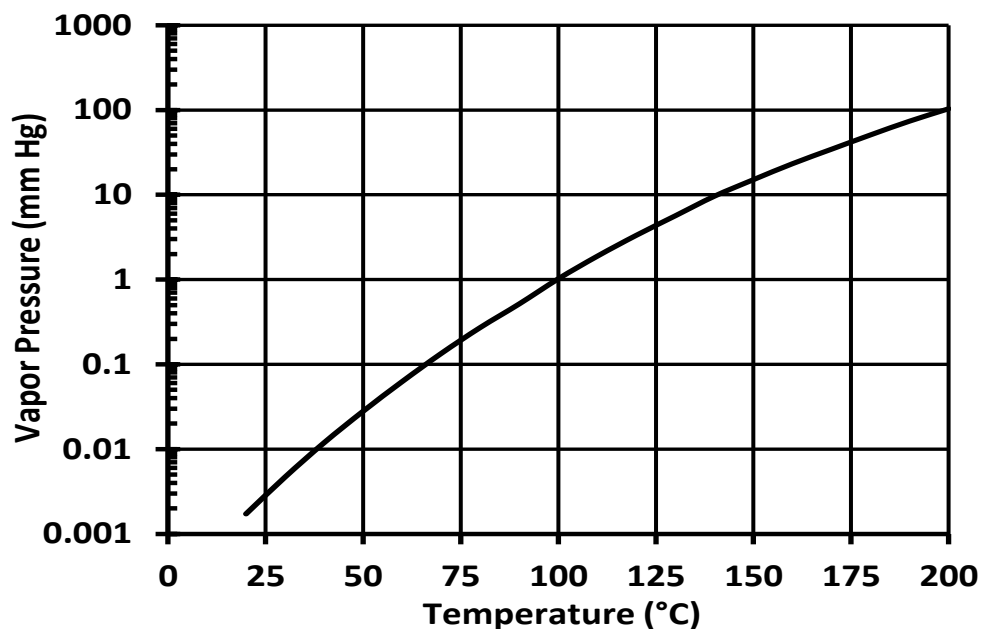
**HEXAHYDROPHTHALIC ANHYDRIDE
(HHPA)
CAS Reg. No. 85-42-7**

Typical Properties

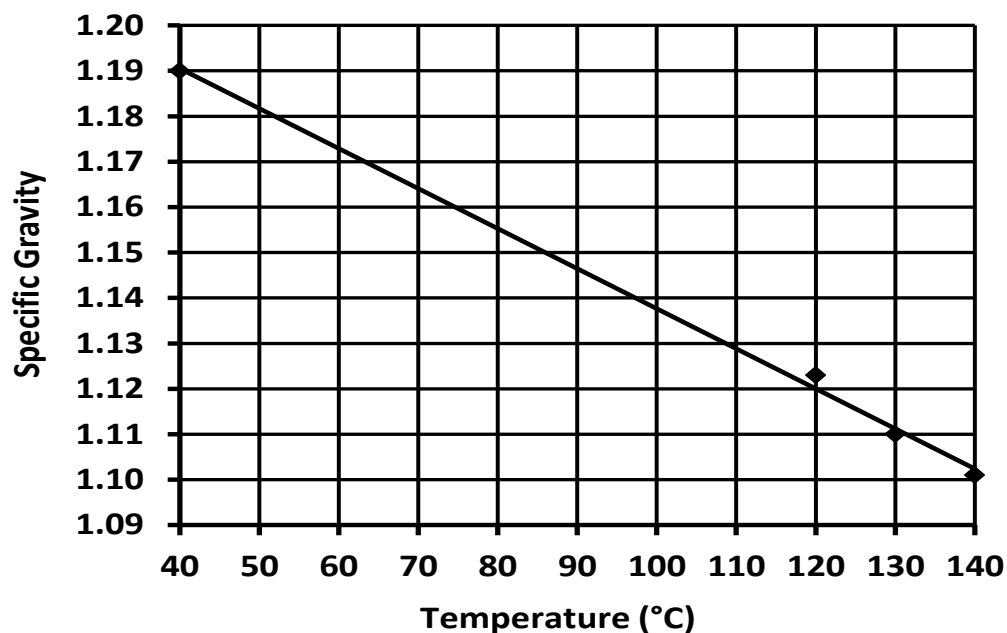
Properties	Values	Test Method
Molecular Weight	154.2	
HHPA, %	99 Min	D-160M
Specific gravity @ 40° C, g/cc	1.19	D-102 (ASTM D 4052)
Color, Pt-Co	25 Max	D-103 (ASTM D 1209)
Appearance	White, low melting solid	D-104 (Visual)
Melting Point (°C)	34-38	D-134
Free Acid, %	0.5 Max	D-160N

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HEXAHYDROPHthalic ANHYDRIDE

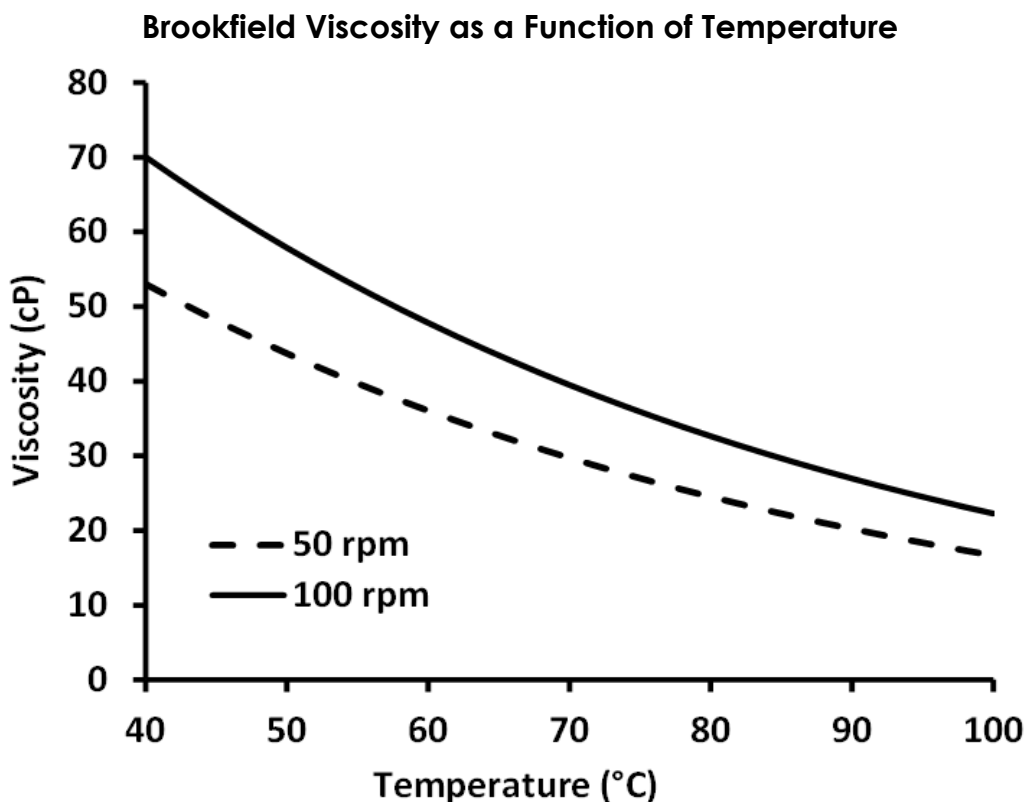
Vapor Pressure as a Function of Temperature



Specific Gravity as a Function of Temperature



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HEXAHYDROPHthalic ANHYDRIDE



Applications

Dixie hexahydrophthalic anhydride (HHPA) is a high purity anhydride product finding use in curing epoxy resins and other polymer applications. Due to its low residual unsaturation level, use of HHPA is preferred over other alicyclic anhydride products when maximum resistance to yellowing in coatings and castings is required, and where premium optical and electrical performance are needed in epoxy casting and encapsulation applications. Low melt viscosity of HHPA and its high mix ratio with epoxy make the product particularly suitable for epoxy applications where high fiber loadings are required. HHPA is commonly used to cure epoxy resins in potting, filament windings, and clear castings.

TECHNICAL BULLETIN

HEXAHYDROPHthalic ANHYDRIDE

For more information on the use of anhydrides like HHPA as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical.

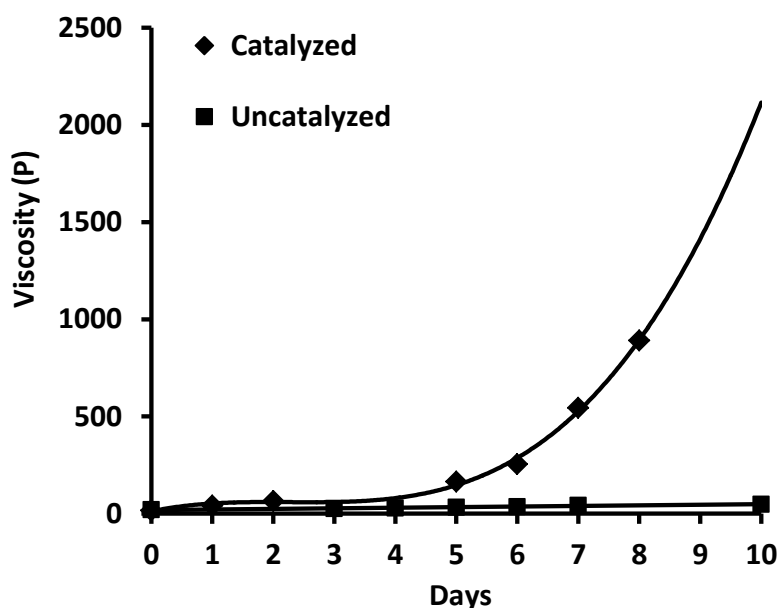
Cure Catalysts

The effect of benzyldimethylamine (BDMA) catalyst on pot life was studied using the following formulations:

System	Uncatalyzed	Catalyzed
EPON Resin 828 (Momentive)	100	100
HHPA	39*	76.7
Benzyldimethylamine (BDMA)	0	1.5

*Epoxy homopolymerization is a significant side reaction for uncatalyzed systems, so anhydride demand is less.

Viscosity as a function of time is summarized in the following plot.



Anhydride Level

The data contained herein are furnished for information only and are believed to be reliable. This information is provided only as guidance and is not to be considered a warranty or quality specification. Dixie Chemical makes no warranties (expressed or implied) as to its accuracy and assumes no liability in connection with any use of this product.

TECHNICAL BULLETIN

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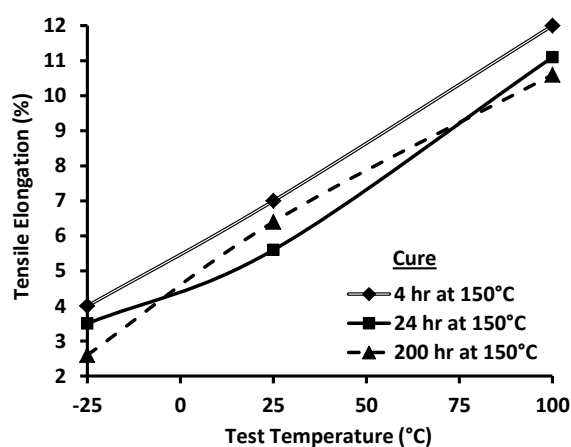
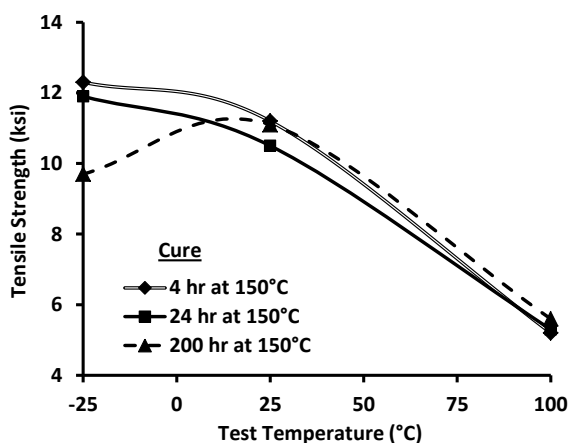
Most applications use between 60-90 phr HHPA with bisphenol A liquid epoxy resins. Most high performance applications where high levels of heat or chemical resistance, or high strength is required utilize 75-90 phr.

Cure Cycle

In one study, 100 parts EPON Resin 828 (Momentive) was combined with 78 parts HHPA and one part BDMA. Samples were cured at 150°C from 4 to 200 hours. Cured specimens were subjected to 24 hours of water boil and 3 hours of acetone boil. The percentage weight gain is summarized below.

Cure (hr at 150°C)	4	24	200
% Wt gain-24 hour water boil	0.45	0.45	0.46
% Wt gain-3 hour acetone boil	1.3	1.3	1.3

Samples cured under the same conditions as those used in the boil test were also tested at -25°C, at room temperature, and at 100°C. The following charts summarize results for tensile strength and elongation:



TECHNICAL BULLETIN

HEXAHYDROPHthalic ANHYDRIDE

Length of cure had no effect on tensile modulus, as the same values were obtained for each post-cure schedule. These results are summarized in the following table:

Test Temperature (°C)	Tensile Modulus (ksi)
-25	400
25	400
100	300

Formulating with Liquid Epoxy Resins

Gel time and exotherm characteristics were determined for 100 parts EPON Resin 828 (Momentive) when combined with 88 parts HHPA and 0.5 part BDMA. Twenty-five gram samples were placed in a bath at 130°C. The following results were obtained:

Minutes from 80°C to gel	4.0
Temperature at Gel (°C)	142
Minutes from 80°C to peak	6.0
Peak Exotherm (°C)	202

Heat distortion temperature and tensile properties were determined for 100 parts EPON Resin 828 combined with 82 parts HHPA and one part BDMA. This system was cured for 2 hours at 90°C, followed by a post-cure of 4 hours at 150°C. The following results were obtained:

Heat Distortion Temp (°C)	130
Tensile Strength (ksi)	450
Tensile Modulus (ksi)	450
Flexural Modulus (ksi)	450

TECHNICAL BULLETIN
HEXAHYDROPHthalic ANHYDRIDE

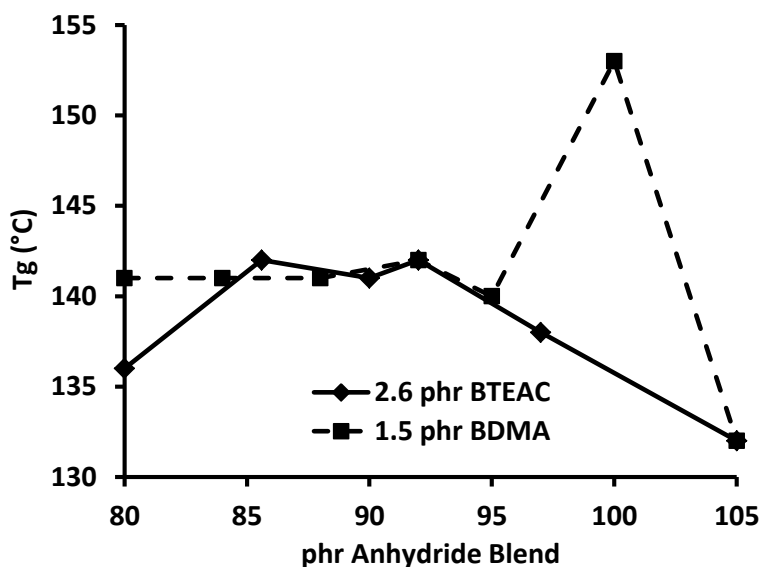
Electrical and mechanical properties were determined for 100 parts EPON Resin 828 combined with 100 parts HHPA and 0.3 part BDMA. The following results were obtained:

Dielectric Constant	
1 Hz	3.32
1 MHz	3.11
Power Factor	
1 Hz	0.0032
1 MHz	0.0148
Heat Distortion Temp (°C)	155
Izod Impact (ft-lb/in)	0.86
Flexural Strength (ksi)	20.8
Flexural Modulus (ksi)	470
% Weight loss - 24 hr at 200°C	0.05

A blend of 80 parts HHPA and 20 parts methylhexahydrophthalic anhydride (MHHPA) was made. Formulations were prepared with bisphenol A liquid epoxy resin at anhydride levels from 80-105 phr. 2.6 phr BTEAC and 1.5 phr BDMA were used as accelerators. Samples were cured for 30 minutes at 100°C, the temperature was raised to 160°C, and cure was continued for 30 minutes. This gave a total cure cycle of about 70 minutes. The following results were obtained for glass transition temperature (T_g):

TECHNICAL BULLETIN

HEXAHYDROPHthalIC ANHYDRIDE



Handling & Storage

HPHA will react with water to form hexahydrophthalic acid. This is normally undesirable, so HHPA should be stored in such a way that it is carefully protected from moisture contamination. The most preferable means of handling HHPA is to store the solid in tightly sealed containers which are carefully protected from moisture contamination. It can then be melted just prior to use. When properly stored, HHPA has a shelf life of 24 months.

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

TECHNICAL BULLETIN

HEXAHYDROPHthalic ANHYDRIDE

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

Hexahydrophthalic anhydride is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.